AVISHKAR 2026

Maharashtra State Inter-University Research Convention

Topic: iTantra, an Offline Neural Speech Transceiver for Low-Bitrate Radio Links

Category: Engineering & Technology Code: SIH26173 Level: UG

Zonal Round

Speech in (any of 10 languages) → **On-device ASR** (sherpa-onnx · 349 MB) → **Text packet** (49 to 315 bytes per message) → **Radio link** (Bluetooth · Wi-Fi Direct fallback) → **On-device TTS** (Piper voices · 7 languages) → **Speech out** (spoken aloud, in order)

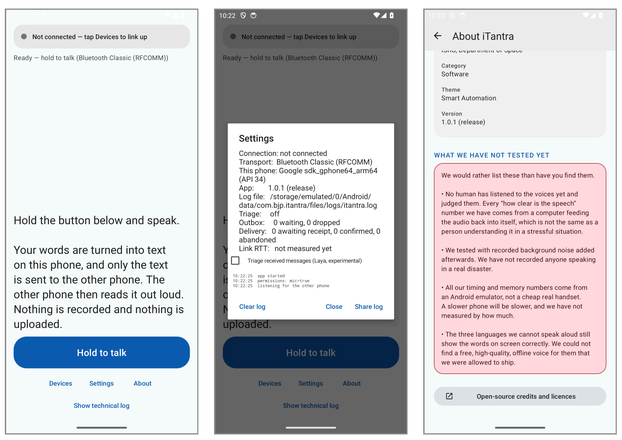
ABSTRACT

Voice carries too much data for a weak radio link. But in a disaster, voice reaches everyone, even people who cannot read. iTantra turns speech into text on the phone. Only the text crosses the link. The other phone speaks it aloud. Recognition runs offline and covers 10 Indian languages. One message is a packet of 49 to 315 bytes over Bluetooth or Wi-Fi Direct. That is about 65 times smaller than the same speech as audio. A delivery layer keeps every message whole when the link drops. It adds an acknowledged outbox, receipts, heartbeats, alert escalation, and survival of an app kill. Mean character error rate is **0.0918** over 2,000 clips. The two-phone loop has run on real phones.

HYPOTHESIS

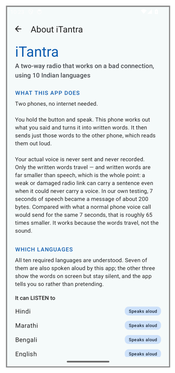
1. Speech recognition on the phone is accurate enough to replace the cloud in an emergency.
2. Text works as the wire format. A few hundred bytes per sentence survive a weak link that audio cannot.
3. Pruning the model output to the ten required scripts removes cross-script noise. The model cannot write characters outside those scripts.
4. Deterministic delivery makes a lossy link behave the same way each time. An acknowledged outbox, receipts, and ordered resends give this result. Alerts preempt normal speech.
5. A killed app resumes pending messages from disk. Nothing is lost in silence.
6. Low-RAM phones can run a 349 MB model if the app returns idle memory under pressure.

EVIDENCE: THE REAL APP



Screens from release v1.0.1 on Android 34: home, settings, and About. The two-phone loop ran on real phones on 27 Sep 2026.

INTRODUCTION



Problem statement SIH26173 asks for a voice radio for weak links. It comes from ISRO, Department of Space. In floods, disasters, and remote posts, the radio link is weak and intermittent. That is exactly when a message must get through. Voice is the natural interface. But raw voice, and even compressed voice, overloads a damaged link and dies on packet loss. Cloud speech services need internet. There is no internet there.

Our approach turns the stack around. The sender phone recognises the speech offline. Only text crosses the radio. Text is small, and the link can order it and acknowledge it. So the link can drop and recover without losing the message. The receiver phone speaks the text aloud. There is no cloud and no account. The app never asks for the INTERNET permission.

FIXED FOR ALL BUILDS

Rationale: only text crosses the link. About 7 seconds of speech become about 200 bytes. That is roughly 65 times smaller than a voice call. We treat reliability as a delivery problem, not a radio problem. Every segment is acknowledged or resent, with a backoff of 3, 6, 12, and 15 seconds and up to 10 tries. Heartbeats go out every 4 seconds. A peer is declared gone after 12 seconds. One configuration ships for every run: arm64-v8a, minSdk 23, a pruned vocabulary of 1,002 tokens, and 96 unit tests on every build.

Run	Configuration and test set	Result
A: baseline	ASR v1, 10,288-token vocabulary, FLEURS clean, 2,000 clips	Mean CER 0.0992
B	ASR v2, FLEURS clean, 2,000 clips	Mean CER 0.0934
C: shipped	ASR v3, pruned 10,288 to 1,002 tokens, FLEURS clean, 2,000 clips	Mean CER **0.0918**. Zero out-of-script characters.
D	C, pink noise at 15 dB SNR, Hindi clips	Script retention 40% to **78%**
E	C, noise at 0 dB SNR, Tamil clips	44% retention. Weakest case, declared.
F: device	Shipped build, on-device self-tests, Android 34, arm64	3 self-tests green. ASR RTF 0.156. TTS RTF 0.078.

REAL CODE: DEAD-LINK DETECTION

`transport/Heartbeat.kt` — SHIPPED

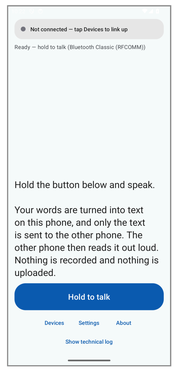
```
class Heartbeat(
    private val intervalMs: Long = PING_INTERVAL_MS,
    private val timeoutMs: Long = DEAD_AFTER_MS,
) {
    companion object {
        const val PING_INTERVAL_MS = 4_000L
        const val DEAD_AFTER_MS = 12_000L
    }
}
```

SYSTEM COMPONENTS

Component	Size / spec	Role
Offline ASR: Omnilingual CTC (int8, pruned vocabulary)	349 MB	speech to text, 10 languages
VAD: Silero	0.63 MB	splits speech into segments, end-of-speech flush
Neural TTS: Piper voices (hi, mr, bn, ta, te, ml, en)	≈60 MB per voice	text to speech, 7 of 10 languages
Wire format: JSON envelope (id, lang, ttl, type)	49 to 315 B	the radio payload
Transport: Bluetooth RFCOMM, Wi-Fi Direct fallback	2 Mbit/s class	link layer
Delivery layer: acknowledged outbox, receipts, ledger, PendingStore	temp-file + rename	order, dedupe, kill survival
Optional triage: Laya int8, off by default	143 MB	flags destructive language as a priority alert

App: Kotlin, arm64-v8a, minSdk 23. Signed release APK v1.0.1, 993 MB, zero network permissions.

RESULTS AND CONCLUSION



Shipped v1.0.1: a 993 MB signed APK for Android 6+ on arm64, with zero network permissions. Measured over 2,000 FLEURS clips in 10 languages (6.7 hours): mean character error rate **0.0918**. The model wrote zero characters outside the ten scripts. Under pink noise, Hindi script retention rose from 40% to **78%**. The script filter dropped foreign-script segments whole in 25 of 25 cases. Hindi speech round-trip error fell from 0.339 to **0.102**. On the phone, recognition runs at a real-time factor of 0.12 to 0.16. Speech synthesis runs at 0.055 to 0.08. Mouth to ear takes about 1 second. The app uses 428 MB of memory with the model loaded.

96 unit tests and three on-device self-tests pass. The two-phone loop ran end to end on real phones on 27 Sep 2026. **Conclusion:** a text-first transceiver makes voice usable over weak, unreliable links. It runs fully offline, on ordinary phones.

FUTURE SCOPE

1. Human listening tests and field trials. The noise sets are synthetic. Real field audio is the next measurement.
2. Latency, RAM, and battery numbers on a real budget handset.
3. Neural voices for the remaining three languages: Gujarati, Kannada, and Odia. They are text-only today.
4. Language identification, to separate Hindi and Marathi, and Bengali and Assamese.
5. A foreground service for very long holds. A smaller APK.

REFERENCE

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5. Microsoft. ONNX Runtime. github.com/microsoft/onnxruntime
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8. iTantra. Release v1.0.1. DELIVERY_FINDINGS and VERIFICATION_STATUS are in the repository. github.com/psb-001/Itantra (2026)

GitHub repo | Release v1.0.1 | Findings | Everything